Adaptive Hierarchical Cue Hashing for Cross-modal Retrieval

Yinglin Wei[1] and Meng Wang[2*]

[1]School of Computer Science and Technology, Guangxi University of Science and Technology, Liuzhou, 545000, China.
[2*]Tus College of Digit, Guangxi University of Science and Technology, Liuzhou, 545000, China.

*Corresponding author(s). E-mail(s): mwang007@gxust.edu.cn;
Contributing authors: 20240701011@stdmail.gxust.edu.cn;

Abstract

Cross-modal hashing has become increasingly important for large-scale multimedia search due to its low storage cost and high retrieval efficiency. Despite the success of large-scale pre-trained vision-language models (VLMs), most existing methods still rely on batch-level softmax-based contrastive losses for image-text alignment. This paradigm restricts effective bidirectional knowledge exchange, limits deep cross-modal interactions, and aggravates the mismatch between different modalities. In addition, VLMs often overemphasize superficial modality-specific patterns, which leads to two key issues: modality imbalance and semantic decay. To address these limitations, we propose Adaptive Hierarchical Cue Hashing (AHCH), a framework that integrates adaptive cue learning with hierarchical hash code generation. First, we leverage cue tuning to extract and compress real-world textual information, thereby enhancing cross-modal representational alignment. We then introduce Adaptive Hierarchical Fusion (AHF), which dynamically integrates multi-level features across layers to maintain consistency in meaning and mitigate inter-modal discrepancies. Furthermore, the Semantic Instance Alignment (SIA) module aligns instances with similar labels while jointly optimizing intra-modal and inter-modal similarity. This design improves the discriminability and compactness of the learned hash codes and helps reduce quantization error. Experiments on three public benchmarks validate AHCH, which effectively enhances cross-modal interaction and generates discriminative hash codes. Code and links to official datasets are available at https://github.com/wyl20000205/AHCH.

Keywords: Cross-modal hashing, Image-text retrieval, Vision-language models, Hamming space

1 Introduction

With the rapid growth of multimedia data, bridging heterogeneous modalities such as images and text has become a core challenge in information retrieval [1, 2]. Traditional cross-modal retrieval methods rely on custom neural architectures to extract real-valued embeddings, where similarity is measured in a shared embedding space [3, 4]. While effective in small-scale to medium-scale settings, these approaches face severe bottlenecks due to high computational cost and limited scalability as data volumes expand. Distributional and representational differences across modalities create a persistent heterogeneity gap, complicating accurate retrieval. To reduce this overhead, cross-modal hashing methods map heterogeneous data into a common Hamming space [5, 6]. Here, data are encoded into compact binary hash codes that reduce storage costs [7, 8]. This mapping is achieved by learning a shared hash function with strong representational capacity. This framework enables efficient similarity computation and considerably accelerates large-scale retrieval, making hashing a widely adopted and fundamental solution for multimodal retrieval tasks [9, 10]. Benefiting from the advancement of large language models, a series of self-supervised models has established a new paradigm for visual and textual representation learning.

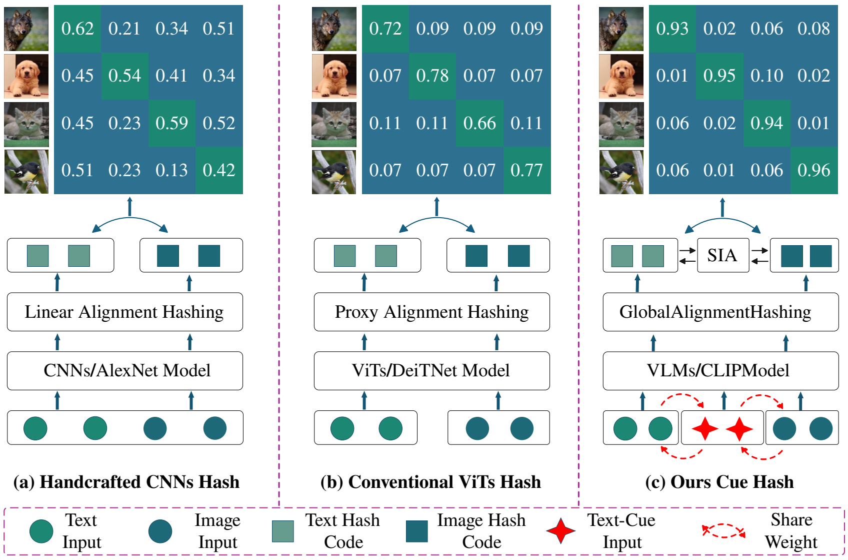

Fig. 1 Illustration of different cross-modal hashing frameworks. The handcrafted CNNs-based method extracts visual features with CNNs and aligns modalities through linear hashing [11, 12], while the transformer-based approach employs ViTs[13] to encode features and relies on proxy hashing for representation alignment [14, 15]. In contrast, our proposed Cue Hash introduces cue-guided inputs with shared weights and integrates a Semantic Interaction Alignment component, enabling more effective cross-modal consistency and finer-grained alignment between image and text representations. The colored blocks at the top of each subfigure represent the corresponding similarity matrices between image and text features.

These models, pre-trained on large-scale multimodal pairs, provide strong cross-modal feature extraction and interaction, supporting both general retrieval and domain-specific vision-language applications such as healthcare [16]. The recently proposed DAGtH [17] method leverages the large-scale pretrained vision-language model CLIP [18, 19] to extract visual and textual features. In the hash learning stage, it employs the deep adaptive gradient-triplet technique to embed heterogeneous modality data into a discriminative discrete space while capturing neighborhood relationships in the original feature space to preserve label information. However, its strong reliance on neighborhood relationships may introduce noise when modality distributions are inconsistent, and it lacks explicit constraints for cross-modal semantic alignment. Similar supervised approaches, such as DADH [12] and DDBH [20], have not explicitly addressed these issues. Moreover, the unsupervised HCH [21] method uses a VLM backbone to extract visual and textual features and enhances multimodal information fusion through a cross-modal Transformer. Nevertheless, its reliance on the pretrained model and insufficient fine-grained awareness hinder the generation of high-quality hash codes. Other unsupervised methods, including KDCMH [22] and CIRH [23], similarly overlook these aspects in complex cross-modal retrieval scenarios.

Based on the above analysis, Fig. 1 shows that current cross-modal hashing (CMH) methods primarily focus on feature learning and hash learning, yet existing approaches struggle to balance the relationship between the two. During feature learning, the key challenge is **modality imbalance**, as VLMs are typically pre-trained once on large-scale image-text pairs and then kept frozen during downstream tasks. For example, CLIP is trained on 400 million image-text pairs, which may lead to modality imbalance and hinder cross-modal alignment in meaning. In the subsequent hash learning stage, the main issue lies in **semantic decay**, where existing methods often model neighborhood relationships only in the original feature space and overlook the dynamic adaptability of cross-modal features. This may introduce noise and prevent the hash codes from capturing fine-grained content information. Consequently, modality alignment in feature learning and representation preservation in hash learning remain the main bottlenecks of CMH methods, especially under diverse downstream distributions and retrieval conditions.To address the above challenges in CMH, we propose Adaptive Hierarchical Cue Hashing (AHCH), as illustrated in Fig. 2, which integrates cue learning, adaptive hierarchical fusion, and semantic instance alignment into a unified deep cross-modal hashing framework. Specifically, the method injects class-conditional cues into both image and text features, forming visual-cue and text-cue representations to enhance feature expressiveness. Subsequently, an adaptive hierarchical fusion network performs multi-level integration across different feature layers of these enhanced representations, further improving their representational capacity and robustness across both modalities. Finally, the Semantic Instance Alignment (SIA) module enhances the conceptual consistency and quality of the generated hash codes by directly optimizing instance-level similarities. The main technical contributions of this paper are summarized as follows:

Table 1 Comparison of AHCH with selected recent CLIP- and VLM-based cross-modal hashing methods.

Method	Prompt or Cue Learning	Hierarchical Fusion	Instance-Level Alignment	Out-of-Domain Evaluation
NSDH [24]	No	No	No	No
DDBH [20]	No	No	No	No
DSSH [25]	No	No	No	No
AHCH	Yes	Yes	Yes	Yes

- Firstly, cue learning is incorporated into the feature learning stage, where generated cue embeddings are embedded into the bidirectional interaction embedding network to extract and compress real-world textual semantics, thereby more effectively enhancing cross-modal representational alignment.
- Secondly, an adaptive hierarchical fusion strategy integrates multi-level features, preserving semantic consistency, reducing cross-modal discrepancies, and improving robustness and generalization.
- Finally, semantic instance alignment optimizes cross-modal matching. By optimizing intra-modal and inter-modal similarities, it generates compact, discriminative hash codes while improving accuracy.

2 Related work

We review supervised and unsupervised cross-modal hashing methods, together with cue learning for VLMs. Accordingly, this study addresses supervised cross-modal hashing with label guidance. Table 1 compares AHCH with selected recent VLM-based hashing methods in cue learning, hierarchical fusion, instance-level alignment, and out-of-domain evaluation. The comparison highlights the components introduced by AHCH and provides an overview of its methodological scope relative to existing approaches.

2.1 Unsupervised Cross-Modal Hashing

Unsupervised cross-modal hashing methods can learn hash codes without label supervision, offering better scalability and adaptability. However, they often struggle to maintain sufficient content-level relevance. To address these challenges, various approaches have been proposed, each with specific limitations. UKD [26] leverages knowledge distillation to improve hash quality but heavily relies on the teacher network's performance. DAEH [27] employs hybrid similarity estimation with an adaptive teacher-guided strategy, yet remains sensitive to distance metric assumptions. CKDH [28] adopts a CLIP-based framework to effectively capture cross-modal correlations, but incurs high computational cost and risks overfitting. UCHM [29] introduces similarity-friendly loss with bit selection module to generate unified hash codes, though it requires careful hyperparameter tuning. More recently, UHDSH [30] constructs a dynamic similarity hypergraph to integrate structural information and relational cues, but the computation is costly and sensitive to noise. Beyond cross-modal hashing, adaptive clustering and weighted-regularization contrastive learning have been employed to improve discriminative representation learning in unsupervised person re-identification [31]. In summary, although these methods have advanced UCMH, achieving stronger relevance and robustness without reducing efficiency remains challenging. Recent dataset distillation methods based on noise-unconstrained generative modeling provide another direction for reducing training data requirements and improving overall data efficiency [32].

2.2 Supervised Cross-Modal Hashing

Supervised cross-modal hashing methods leverage label information to improve the consistency and reliability of the learned hash codes. These methods utilize labeled multimodal data to establish accurate cross-modal associations and enhance retrieval accuracy through explicit label guidance. Representative SCMH approaches include AGAH [11], which introduces an asymmetric hashing strategy with multi-label binary mapping and an adversarial multi-label attention module, though its ranking-based similarity learning may yield suboptimal hash correlations. DADH [12] employs deep adversarial discrete hashing to reduce distribution gaps across modalities but insufficiently exploits label-driven information. NSDH [24] integrates label matrices with similarity information to strengthen representation quality yet may overlook complex nonlinear relationships. LPGOH [33] adopts label-propagation-based graph optimization to model label dependencies but incurs high computational cost and is sensitive to sparse or noisy annotations. Overall, SCMH methods improve cross-modal relevance and hash code reliability, yet challenges remain in expressive capability, particularly when semantic labels are incomplete, imbalanced, or noisy.

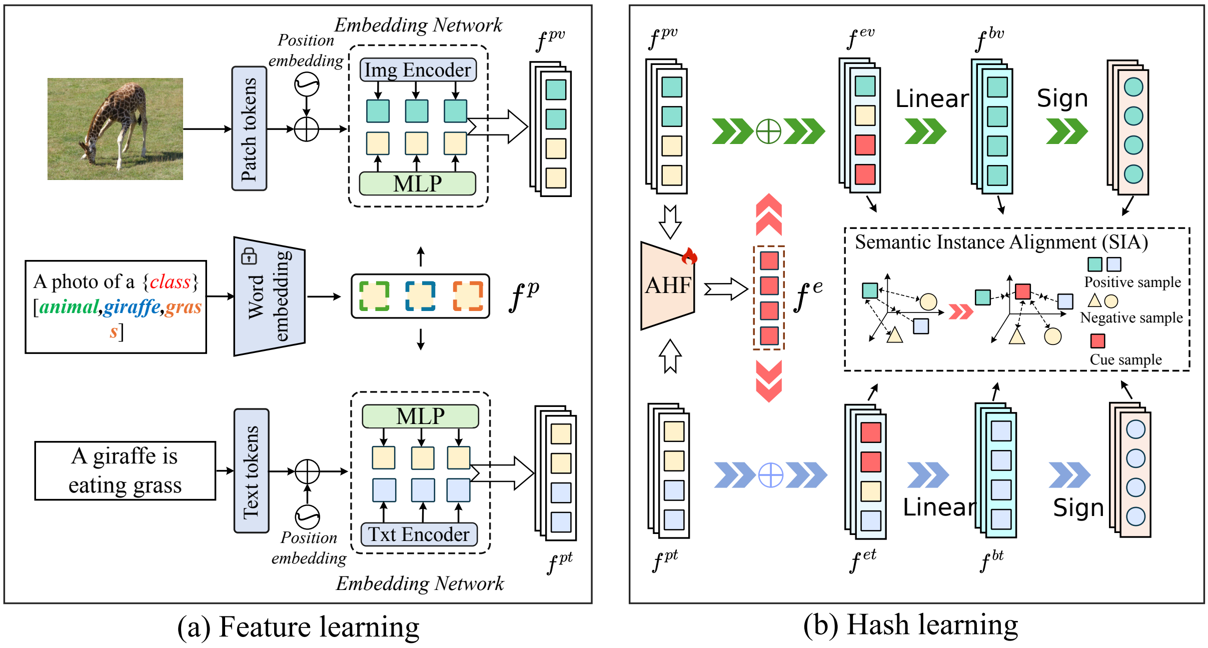

Fig. 2 AHCH has two stages. *(1) Feature learning stage*: adaptive cue embeddings enter the bidirectional interaction embedding network to effectively compress fine-grained textual details, thereby enhancing cross-modal representational alignment. *(2) Hash learning stage*: adaptive hierarchical fusion (AHF) integrates multi-level features to maintain cross-modal consistency and reduce modality gaps, whereas semantic instance alignment (SIA) optimizes intra-modal and inter-modal relationships. Finally, compact, discriminative binary codes are obtained through transformation and quantization.

2.3 Cue Learning for Vision-Language Models

Cue learning originated in natural language processing, driven by efforts to investigate how pre-trained models such as BERT [34] or GPT [35] adapt to downstream tasks without extensive parameter tuning. Typically, cloze-style tasks are constructed, e.g., "No reason to watch. It was [MASK]," where the model infers masked content. In cat-versus-dog classification, the masked token becomes "cat" or "dog." The core idea is designing [MASK] in formats aligned with model pre-training, referred to as cue templates. Compared with conventional cross-modal hashing methods that rely primarily on coarse category-level labels (e.g., "cat"), the resulting representations often fail to capture fine-grained semantic variations and thus miss many relevant instances. In contrast, AHCH conditions the hashing process on richer cue-based textual descriptions that encode multiple topical attributes (e.g., object category, color, and scene), enabling more accurate and efficient cross-modal retrieval. Fine-grained lexical-semantic modeling has also been shown to improve semantic coherence in video paragraph captioning, highlighting the importance of detailed textual representations in broader vision-language learning tasks [36].

A major advantage of cue learning in VLMs is that the vision-language model can learn a set of optimizable visual cues, each associated with a key, and select the most relevant one by computing the cosine similarity between the [CLASS] token and the corresponding key. Unlike manually constructed cues, cue learning automatically selects cues during fine-tuning, converting contextual words into learnable context vectors. These vectors utilize [TOKEN] embeddings at the input and [EOS] embeddings at the output, enabling the optimization of continuous cues rather than relying on fixed manual templates. This design helps mitigate catastrophic forgetting and leverages the model's intrinsic knowledge for efficient knowledge transfer and robust task generalization. Related multimodal research has more broadly explored spatio-temporal disentanglement and constrained self-attention to model modality-specific dynamics and cross-modal dependencies [37]. Building on this idea, AHCH extends cue learning to VLMs, focusing on embedding-level optimization by converting static text cues into learnable continuous vectors. This study introduces cue learning to cross-modal hashing, systematically evaluating its cross-task transferability and representational generalization under diverse downstream retrieval settings across datasets.

Table 2 The definitions of the feature and hash symbols.

Symbol(s)	The description
f^v/f^t	Visual / textual feature vector.
f^p	Class-conditional cue feature.
f^{pv}/f^{pt}	Cue-enhanced visual / textual feature.
f^e	Fused multi-level (modality-agnostic) feature.
f^{ev}/f^{et}	Enhanced visual / textual feature.
f^{bv}/f^{bt}	Real-valued visual / textual hash feature.
b^v/b^t	Visual / textual binary hash code.
$\Theta^v_{ij}/\Theta^t_{ij}$	Visual / textual intra-modal similarity.
Ω_{ij}/Φ_{ij}	Visual-textual cross-modal similarity

3 Proposed method

3.1 Notation

We consider a training set of N image-text pairs $\mathcal{D} = \{d_i\}_{i=1}^N$. Each sample is represented as $d_i = \{v_i, t_i, l_i\}$, where $v_i \in \mathbb{R}^{d_v}$ is the i-th raw image, $t_i \in \mathbb{R}^{d_t}$ is its associated text description, and $l_i \in \{0,1\}^{C\times 1}$ is a multi-hot label vector over C categories. If the i-th sample belongs to the j-th category, then $l_i^j = 1$; otherwise, $l_i^j = 0$. Many vision-language benchmarks are multi-label, and this formulation naturally accommodates such settings. To capture label-level relationships between instances, we construct a binary similarity matrix $\mathcal{S} = \{S_{ij}\} \in \{0,1\}^{N\times N}$, where $S_{ij} = 1$ if d_i and d_j share at least one common label, and $S_{ij} = 0$ otherwise. Later, when constructing class-conditional cues (Sec. 3.2), we derive a single-label index p_i from l_i by selecting its dominant category. The goal of cross-modal hashing is to learn modality-specific hash functions $b^v = H^v(v_i;\psi^v) \in \{+1,-1\}^K$ and $b^t = H^t(t_i;\psi^t) \in \{+1,-1\}^K$ that map an image v_i and a text t_i into compact K-bit binary codes for the visual and textual modalities, respectively, where ψ^v and ψ^t are learnable parameters **during training**. The Hamming distances between these codes are expected to preserve the label-based semantic similarity encoded in $\mathcal{S}$, i.e., semantically similar pairs should have small Hamming distances and dissimilar pairs should be far apart.

We summarize the main notations for features and hash codes in Table 2, including visual and textual features f^v, f^t, the cue feature f^p, cue-enhanced features f^{pv}, f^{pt}, fused and enhanced features f^e, f^{ev}, f^{et}, the real-valued hash features f^{bv}, f^{bt}, and the final binary hash codes b^v, b^t. Throughout, we use a hat $\hat{(\cdot)}$ or tilde $\tilde{(\cdot)}$ to denote intermediate transformed features when needed for clarity.

3.2 Feature Learning

In this work, we build on the CLIP framework with a ViT-B/32 backbone to extract modality-specific features for cross-modal retrieval. Following standard practice, we use the CLIP visual encoder $E_V(\cdot;\theta_v)$ and text encoder $E_T(\cdot;\theta_t)$ as backbone encoders and keep their parameters frozen during training, while learning the cue module, the Adaptive Hierarchical Fusion (AHF) module, and the hash heads.

3.2.1 Class-conditional textual template

To generate cue-aligned features, we design a class-specific textual template of the form t_c = "a photo of a [CLASS]". For each image-text pair (v_i, t_i) with multi-label annotation l_i, we derive a single-label index p_i by selecting its primary category:

$$p_i = \arg\max_{1\le c\le C} l_{i,c} \tag{1}$$

The placeholder [CLASS] in the template t_c is then replaced with the textual name of the class indicated by index p_i yielding a class-conditional sentence. The resulting template is tokenized as $\text{text\_cue}_i = \text{tokenizer}(t_c)$, and converted into a continuous embedding space via a learnable cue module:

$$\hat{f}^p = \text{Embedding}(\text{text\_cue}_i) \quad f^p = \text{MLP}(\hat{f}^p) \tag{2}$$

Here, Embedding$(\cdot)$ denotes a trainable token embedding layer and MLP$(\cdot)$ is a lightweight feed-forward network. The resulting f^p serves as a class-conditional semantic prototype that will be shared by both visual and textual branches.

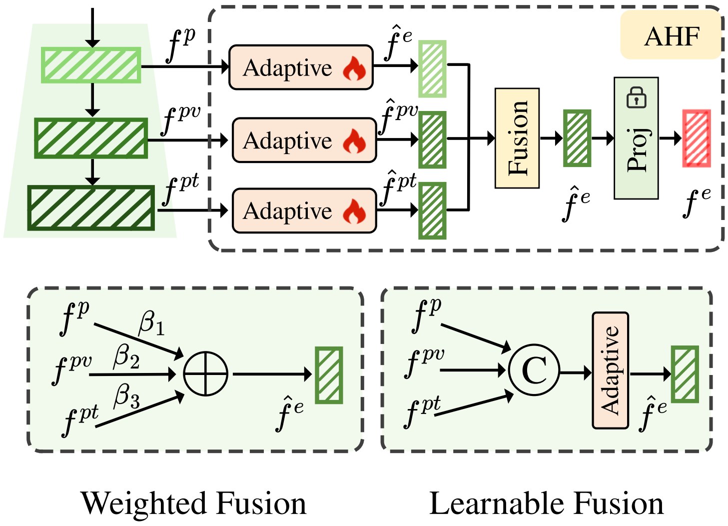

Fig. 3 Adaptive Hierarchical Fusion (AHF) provides two fusion branches to enhance feature integration. *(1) Weighted Fusion (WF):* Performs weighted summation of input features, prioritizing their spatial distribution. *(2) Learnable Fusion (LF):* Employs a small parameterized network to linearly merge features and learn complex interaction patterns.

3.2.2 Backbone feature extraction

Given the raw image v_i and text t_i, CLIP encoders first produce high-level modality-specific representations, which are then further projected to obtain f^v and f^t:

$$\hat{f}^v = F_V(E_V(v_i;\theta_v)), \quad \tilde{f}^v = \mathrm{LayerNorm}(\hat{f}^v), \quad f^v = F_H^V(\tilde{f}^v) \tag{3}$$

$$\hat{f}^t = F_T(E_T(t_i;\theta_t)), \quad \tilde{f}^t = \mathrm{LayerNorm}(\hat{f}^t), \quad f^t = F_H^T(\tilde{f}^t) \tag{4}$$

where $F_V(\cdot)$ and $F_T(\cdot)$ are modality-specific projection layers, and $F_V^H(\cdot)$, $F_T^H(\cdot)$ are projection heads that produce final visual and textual features. The image encoder $E_V(\cdot)$ partitions each image into a sequence of patches and augments them with a class token, while the text encoder $E_T(\cdot)$ maps the sentence into token-level embeddings. Layer normalization stabilizes training by re-scaling intermediate features.

3.2.3 Cue-enhanced modality features

To inject class-level semantics into both modalities, we introduce two lightweight branch-specific gating modules that adaptively combine the cue feature with the corresponding modality feature. For the visual branch, the instance-wise coefficients are independently computed for each image-text training pair as

$$[\alpha_p^v, \alpha_v] = \mathrm{softmax}\left(W_v[f^p; f^v]\right), \qquad f^{pv} = \alpha_p^v f^p + \alpha_v f^v. \tag{5}$$

Similarly, the textual branch independently computes

$$[\alpha_p^t, \alpha_t] = \mathrm{softmax}\left(W_t[f^p; f^t]\right), \qquad f^{pt} = \alpha_p^t f^p + \alpha_t f^t. \tag{6}$$

Here, $[\cdot;\cdot]$ denotes vector concatenation, while W_v and W_t are branch-specific learnable projection matrices. The coefficients satisfy $\alpha_p^v + \alpha_v = 1$ and $\alpha_p^t + \alpha_t = 1$ for the visual and textual branches, respectively. Assuming $f^p, f^v, f^t \in \mathbb{R}^d$, their concatenated features belong to $\mathbb{R}^{2d}$, $W_v, W_t \in \mathbb{R}^{2\times 2d}$, and $f^{pv}, f^{pt} \in \mathbb{R}^d$. The cue-enhanced features adaptively balance class-level cues and instance-level modality information before subsequent refinement by the AHF module at different hierarchical levels of the network.

3.3 Adaptive Hierarchical Fusion

We now introduce the Adaptive Hierarchical Fusion (AHF) module, which is designed to fully exploit multi-level features obtained from the embedding network and cue module. As illustrated in Fig. 3, AHF treats features from different stages: cue-enhanced features f^{pv}, f^{pt} and the class-conditional cue feature f^p as complementary embeddings spanning from fine-grained details to higher-level semantics. By adaptively transforming and fusing these embeddings, AHF preserves local details from lower layers while leveraging abstract semantic cues from higher layers.

3.3.1 Adapters for multi-level feature refinement

We first refine the three input features f^{pv}, f^{pt} and f^{p} using separate adapter modules:

$$\tilde{f}^{*} = \text{Adapter}(f^{*}), \quad * \in \{pv, pt, p\} \tag{7}$$

Each adapter captures both local correlations and long-range dependencies within the feature, and is implemented as:

$$\text{Adapter}(x) = x + \text{MLP}(\text{LN}(x) + \text{LogAttn}(\text{LN}(x))) \tag{8}$$

where $\text{LN}(\cdot)$ denotes layer normalization, $\text{MLP}(\cdot)$ is a small feed-forward network, and LogAttn $(\cdot)$ is a log-scaled attention block described below. The residual connection facilitates gradient flow and encourages the adapter to learn task-specific refinements on top of the backbone features.

3.3.2 Log-scaled attention

To highlight salient components while remaining robust to varying sequence lengths, we contrast our log-scaled attention with a standard softmax attention. Given a sequence of scores $\{x_j\}_{j=1}^{n}$, conventional attention is defined as

$$\text{Attn}(x_i) = \frac{\exp(x_i/\tau)}{\sum_{j=1}^{n}\exp(x_j/\tau)} \tag{9}$$

where $\tau > 0$ is a temperature. As the sequence length n increases, the maximum attention weight can rapidly decay, which may hinder the model's ability to focus on the most relevant elements in long sequences. To alleviate this issue, we introduce Log-scaled Attention, which rescales the logits by a factor depending on the sequence length:

$$\text{LogAttn}(x_i) = \frac{\exp\left((s \cdot \log n) \cdot x_i/\tau\right)}{\sum_{j=1}^{n}\exp\left((s \cdot \log n) \cdot x_j/\tau\right)} \tag{10}$$

where $s > 0$ stabilizes the maximum attention weight, τ is the temperature, and n denotes the sequence length. By increasing the effective sharpness as n grows, Log-scaled Attention mitigates the decay of the maximum attention weight and yields a sharper, more stable distribution, which empirically leads to more robust feature integration for long sequences.

3.3.3 Weighted and learnable fusion of multi-level features

The transformed features $\widehat{f}^{pv}$, $\widehat{f}^{pt}$ and $\widehat{f}^{p}$ are then aggregated using two complementary strategies. **In Weighted Fusion (WF).** WF assigns instance-wise coefficients to each of the three transformed features and computes a weighted sum:

$$[\beta_1^i, \beta_2^i, \beta_3^i] = \text{softmax}(W_f[\hat{f}^{pv}; \hat{f}^{pt}; \hat{f}^{p}]), \quad \widehat{f}^{e} = \sum_{k=1}^{3} \beta_i^k \hat{f}^{k} \tag{11}$$

where W_f is a learnable projection matrix and $\widehat{f}^{(1)}, \widehat{f}^{(2)}, \widehat{f}^{(3)}$ denote $\widehat{f}^{pv}$,$\widehat{f}^{pt}$ and $\widehat{f}^{p}$, respectively. These coefficients are dynamically predicted for each instance, enabling AHF to adaptively emphasize cue-based semantics or modality-specific information. **In Learnable Fusion (LF)**. In parallel, we consider a learnable fusion strategy that concatenates the three transformed features along the channel dimension and refines the result with an adapter (Eq. (8)):

$$\widehat{f}^{e} = \text{Adapter}(\text{Concat}(\widehat{f}^{pv}; \widehat{f}^{pt}; \widehat{f}^{p})) \tag{12}$$

This learnable fusion can capture higher-order interactions among the three feature types. In the full AHCH model we employ LF by default, while WF is used in ablation studies to isolate the effect of the adapter-based fusion. A frozen linear projection layer derived from the ViT architecture then maps $\widehat{f}^{e}$ into a fused multi-level feature f^{e}. To obtain modality-specific enhanced features, we use residual concatenation:

$$f^{ev} = \text{Concat}(f^{pv}; f^{e}), \quad f^{et} = \text{Concat}(f^{pt}; f^{e}) \tag{13}$$

Here, f^{e} serves as a modality-agnostic representation that is shared by both branches, while f^{pv} and f^{pt} retain modality-specific details.

3.3.4 Hash feature extraction and binarization

The enhanced features are further processed by a small hash head to obtain real-valued hash features:

$$f^{bv} = \mathrm{SiLU}(\mathrm{MLP}(\mathrm{BatchNorm}(f^{ev}))), \quad f^{bt} = \mathrm{SiLU}(\mathrm{MLP}(\mathrm{BatchNorm}(f^{et}))) \tag{14}$$

where the MLPs for the two modalities share the same architecture (but not necessarily parameters), and SiLU is used as a smooth bounded activation. Finally, the modality-specific projection functions $H_v(\cdot)$ and $H_t(\cdot)$ map these real-valued hash features into binary codes via a tanh-sign binarization:

$$b^v = \mathrm{sign}(\tanh(H^v(f^{bv}; \psi^v))); \quad b^t = \mathrm{sign}(\tanh(H^t(f^{bt}; \psi^t))) \tag{15}$$

Here, $\tanh(\cdot)$ squashes activations toward ± 1, which helps reduce quantization error, and $\mathrm{sign}(\cdot)$ produces the final $\{+1, -1\}$ hash codes with $\mathrm{sign}(x) = +1$ when $x \geq 0$ and -1 otherwise. During training, we adopt a straight-through estimator to approximate the gradient of the sign function, while at inference time we apply the exact sign operation. This design effectively preserves local details and global semantics, and the residual connections with normalization further stabilize optimization. The AHF module is modality-agnostic and particularly effective when combined with cue-based feature learning for cross-modal retrieval.

3.4 Hash Learning

To preserve intra- and inter-modal semantic correlations, enhance cross-modal consistency, and obtain quantization-friendly hash codes, we propose a Semantic Instance Alignment (SIA) objective. SIA comprises three components: *(1) intra-modal structure preservation*, *(2) cross-modal alignment*, and *(3) quantization regularization*. They encourage the real-valued hash features to form semantically meaningful structures both within and across modalities, while remaining close to their binary counterparts.

3.4.1 Intra-modal structure preservation

Within each modality, instances that share labels should be mapped to similar hash features, whereas semantically dissimilar instances should be pushed apart. We adopt a pairwise logistic-likelihood loss on the real-valued hash features. The intra-modal similarities for images and texts are defined as:

$$\Theta^v_{ij} = \frac{1}{2}\left(f^{bv}_i\right)^T f^{bv}_j, \quad \Theta^t_{ij} = \frac{1}{2}\left(f^{bt}_i\right)^T f^{bt}_j \tag{16}$$

Let B denote the number of paired image-text instances in each mini-batch. In the following losses, $i, j \in \{1, \ldots, B\}$, and S_{ij} denotes the label-based similarity between the i-th and j-th instances within the current mini-batch. Using the label-based similarity matrix S introduced in Sec. 3.1, we define the intra-modal losses for the visual and textual modalities as

$$\mathcal{J}^v_{intra} = \frac{1}{B^2}\sum_{i=1}^{B}\sum_{j=1}^{B}\left[\log\left(1 + e^{\Theta^v_{ij}}\right) - \Theta^v_{ij} S_{ij}\right] \tag{17}$$

$$\mathcal{J}^t_{intra} = \frac{1}{B^2}\sum_{i=1}^{B}\sum_{j=1}^{B}\left[\log\left(1 + e^{\Theta^t_{ij}}\right) - \Theta^t_{ij} S_{ij}\right] \tag{18}$$

These terms encourage hash features of similar instances ($S_{ij} = 1$) to have large inner products, while penalizing large similarities between dissimilar instances ($S_{ij} = 0$) The overall intra-modal structure preservation loss is given by:

$$\mathcal{J}_{intra} = \mathcal{J}^v_{intra} + \mathcal{J}^t_{intra} \tag{19}$$

3.4.2 Cross-modal semantic alignment

To explicitly align semantic relationships across modalities, we further require image-text pairs with the same labels to be close in the joint hash space. We define two cross-modal similarity measures:

$$\Omega_{ij} = \frac{1}{2}\left(f^{bv}_i\right)^T f^{bt}_j, \quad \Phi_{ij} = \frac{1}{2}\left(f^{bt}_i\right)^T f^{bv}_j \tag{20}$$

corresponding to the image-to-text and text-to-image directions, respectively. Using the same label-based similarity matrix S, we formulate a bidirectional logistic-likelihood loss for cross-modal alignment:

$$\mathcal{J}_{inter} = \frac{-1}{B^2} \sum_{i=1}^{B} \sum_{j=1}^{B} \sum_{p \in \{\Omega_{ij}, \Phi_{ij}\}} \left(p \cdot \mathcal{S}_{ij} - \log\left(1 + e^{p}\right)\right) \tag{21}$$

This term encourages paired image-text instances with shared labels to have large cross-modal similarities in both directions and discourages high similarities for mismatched pairs. Using the same $\mathcal{S}$ for intra- and inter-modal terms ensures consistent semantic supervision across modalities during training.

3.4.3 Quantization regularization

To obtain compact and reliable Hamming codes, the continuous hash features should lie close to their discrete counterparts. We therefore impose a quantization regularization during model training:

$$\mathcal{J}_{quan} = \frac{1}{KB} \left(\sum_{i=1}^{B} \|b_i^v - f_i^{bv}\|_2^2 + \sum_{i=1}^{B} \|b_i^t - f_i^{bt}\|_2^2 \right) \tag{22}$$

where K is the hash code length. It penalizes discrepancy between real-valued and binary representations, stabilizing the binarization process and reducing quantization error. In combination with the straight-through estimator in Eq. (15), this encourages the continuous features to approach $\{+1, -1\}^K$.

3.4.4 Semantic Instance Alignment objective

Finally, the overall SIA objective combines the cross-modal alignment, intra-modal structure preservation, and quantization regularization terms within a unified joint optimization framework:

$$\mathcal{J}_{\mathrm{SIA}} = \alpha \mathcal{J}_{inter} + \beta \mathcal{J}_{intra} + \lambda \mathcal{J}_{quan} \tag{23}$$

where α, β, and λ are non-negative hyperparameters that balance the contributions of cross-modal alignment, intra-modal structure preservation, and quantization, respectively. By optimizing $\mathcal{J}_{SIA}$ jointly with the network parameters during training, AHCH learns modality-specific hash functions that produce compact, discriminative, and semantically aligned binary codes for cross-modal retrieval.

4 Experiments

4.1 Dataset Descriptions

(1) MIRFLICKR25K [38] serves as a recognized benchmark for cross-modal retrieval, comprising 24,581 image-text pairs distributed over 24 categories, with each pair annotated with at least one label. Its well-structured annotations and manageable scale make it a standard choice for studies on image-text retrieval, multimodal learning, and cross-modal analysis. **(2) NUSWIDE** [39] represents a large-scale and diverse cross-modal dataset that has been employed in deep hashing research. Following the experimental protocol, categories with insufficient samples were removed during preprocessing, resulting in a processed version containing 195,834 image-text pairs across 21 categories. **(3) MSCOCO** [40] provides a widely used multi-label dataset designed for object detection, covering 80 categories. The dataset is divided into a training set with 82,785 images and a validation set with 40,504 images, where each image receives five captions. In our experiments, the training and validation sets are merged, treating each instance as an image-text pair, with every sample associated with at least one category.

For fairness and consistency, we adopt a unified sampling strategy across all three datasets. Specifically, 5,000 image-text pairs are randomly selected from each dataset to form the query set; the remaining samples of the same dataset constitute the known retrieval database (D_k), and an additional 10,000 pairs are randomly drawn from these remaining samples for model training. The entire samples of the other two datasets are combined to construct the unknown retrieval database (D_{unk}), which is used to assess generalization to unseen data under identical experimental conditions. For uniform input processing, all images are resized to 224×224 pixels and all texts are represented with Byte Pair Encoding (BPE) [41].

Table 3 Comparison of mAP of different baseline methods on three datasets under 16-, 32-, and 64bits hash codes.

Task	Method	MIRFLICKR25K			NUSWIDE			MSCOCO		
		16bits	32bits	64bits	16bits	32bits	64bits	16bits	32bits	64bits
I2T	AGAH	0.7923	0.7945	0.8069	0.6455	0.6600	0.6512	0.5842	0.6314	0.6478
	DADH	0.8020	0.8072	0.8179	0.6492	0.6662	0.6664	0.6233	0.6458	0.6498
	GCDH	0.7910	0.8064	0.8127	0.6483	0.6822	0.6960	0.6832	0.7089	0.7380
	MITH	0.8412	0.8645	0.8718	0.7004	0.7140	0.7297	0.7039	0.7359	0.7664
	DSPH	0.8129	0.8482	0.8541	0.6830	0.6979	0.7162	0.7044	0.7510	**0.7994**
	RDPH	0.7420	0.7749	0.7867	0.6330	0.6317	0.6523	0.5867	0.7151	0.7368
	DDBH	0.8450	0.8534	0.8610	0.7041	0.7145	0.7229	0.7165	0.7454	0.7681
	DSSH	0.8312	0.8419	0.8433	0.7131	0.7248	0.7297	0.7281	0.7322	0.7431
	AHCH	**0.8712**	**0.8841**	**0.8998**	**0.7429**	**0.7482**	**0.7673**	**0.7514**	**0.7600**	0.7832
T2I	AGAH	0.7887	0.7904	0.8049	0.6313	0.6422	0.6336	0.6049	0.6125	0.6423
	DADH	0.7920	0.7959	0.8064	0.6501	0.6679	0.6808	0.6076	0.6270	0.6336
	GCDH	0.7804	0.7945	0.8001	0.6815	0.7376	0.7402	0.6810	0.7131	0.7318
	MITH	0.8228	0.8389	0.8468	0.7140	0.7276	0.7401	0.7017	0.7358	0.7661
	DSPH	0.8000	0.8238	0.8294	0.6997	0.7153	0.7304	0.7040	**0.7556**	0.7713
	RDPH	0.7200	0.7504	0.7713	0.6732	0.7067	0.7190	0.5925	0.6910	0.7139
	DDBH	0.8245	0.8318	0.8390	0.7194	0.7211	0.7325	0.7167	0.7394	0.7595
	DSSH	0.8211	0.8367	0.8393	0.7031	0.7312	0.7356	0.7132	0.7259	0.7493
	AHCH	**0.8675**	**0.8619**	**0.8871**	**0.7364**	**0.7411**	**0.7589**	**0.7407**	0.7533	**0.7724**

4.2 Experimental Details

Our proposed AHCH method is implemented on the PyTorch framework and trained on a workstation equipped with a single NVIDIA RTX 4090 GPU. We adopt the *stochastic gradient descent* (SGD) optimizer with an initial learning rate of $1e^{-5}$. The network is trained for 30 epochs with a batch size of 64. For fair comparison, we employ standard data augmentation strategies, including random resized cropping and horizontal flipping, and the input resolution is set to 224 for all datasets. In AHCH, three balancing hyperparameters α , β, and λ are introduced to regulate different components of the objective function. Specifically, $\{\alpha, \beta, \lambda\}$ are set to $\{0.5, 0.5, 5\}$ in our experiments, while for the MSCOCO dataset with long captions they, are set to $\{0.1, 0.1, 1\}$.

In our experiments, we compared seven state-of-the-art deep cross-modal hashing methods, including AGAH [11], DADH [12], GCDH [42], MITH [43], DSPH [44], RDPH [45], DDBH [20], and DSSH [25]. All methods were implemented using the officially released codes, and their parameters were configured according to the original publications. In all main tables, we report the results of AHCH with the LF fusion branch. The WF variant is only used in ablation studies as a parameter-efficient baseline.

The experiments involve two cross-modal retrieval tasks: image-to-text retrieval (I2T) and text-to-image retrieval (T2I). Following DDBH [20], we adopt five evaluation metrics to provide a comprehensive assessment of retrieval performance: *(1) mean Average Precision (mAP)* under the Hamming ranking protocol, serving as the primary indicator of overall retrieval performance; *(2) Precision-Recall (PR) curves* under the hash lookup protocol, providing comparative insights across different recall levels; *(3) TopN-Precision (TopN-P)* curves, reflecting retrieval accuracy at varying retrieval depths; *(4) NDCG@1000*, a ranking-based metric that considers both the relevance and position of retrieved results, thereby evaluating how well the top-1000 ranking aligns with the ground truth; *(5) NDCG@1000 across different VLMs variants*, including ALIP [46], BLIP [47], and SLIP [48]. This metric measures how different VLM backbone networks affect the overall performance of our method. By comparing results on D_k and the D_unk, it also evaluates the model's ability to generalize to both seen and unseen data.

4.3 Comparison with the Baselines

We have conducted comprehensive comparisons with a series of latest baseline methods on three benchmark datasets. From the results, the following observations can be drawn: (1) As shown in Table 3, AHCH

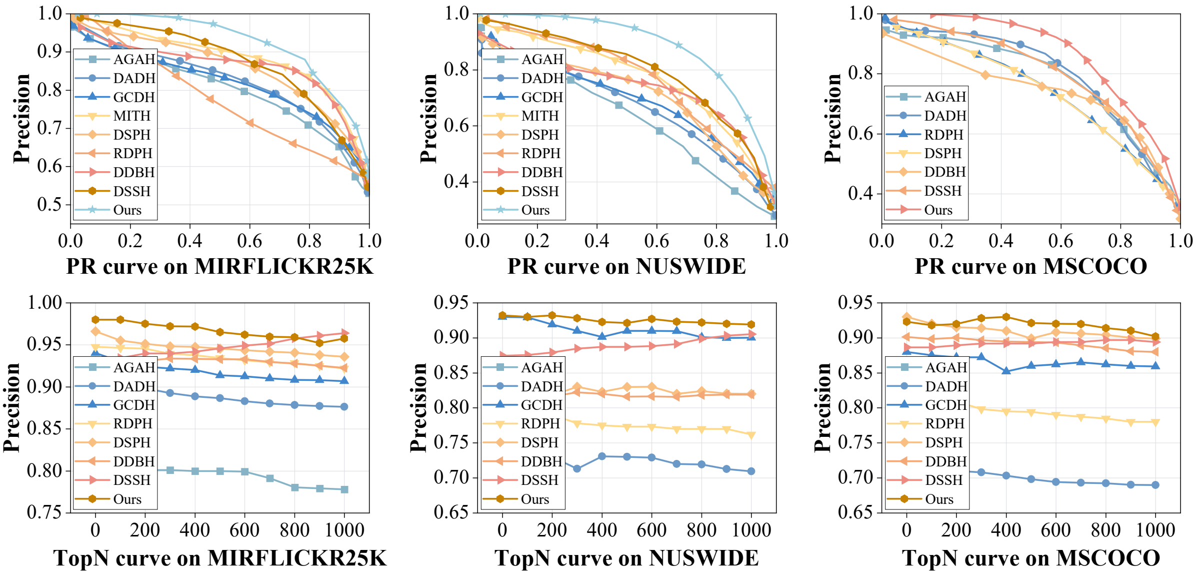

Fig. 4 I2T precision-recall and Top-N curves on MIRFLICKR25K, NUSWIDE, and MSCOCO at 64 bits.

Table 4 Evaluation of the Proposed AHCH Method in Terms of NDCG@1000 When Combined with Different Pre-trained VLMs on F25K (MIRFLICKR25K) under the Same Cross-Modal Retrieval Evaluation Settings.

Task	Method	F25K on *D_k*			F25K on *D_unk*			F25K on *D_k* ∪ *D_unk*		
		16bits	32bits	64bits	16bits	32bits	64bits	16bits	32bits	64bits
I2T	AHCH+ALIP	0.5521	0.5411	0.5692	0.3124	0.3321	0.3351	0.3217	0.3387	0.3291
	AHCH+BLIP	**0.6721**	**0.6701**	**0.6978**	0.3423	0.3351	0.3481	0.3654	0.3714	0.3712
	AHCH+SLIP	0.6231	0.6413	0.6491	0.3223	0.3198	0.3281	0.3792	0.3676	0.3821
	AHCH+CLIP	0.6022	0.6166	0.6091	**0.3942**	**0.4041**	**0.4102**	**0.4121**	**0.4122**	**0.4492**
T2I	AHCH+ALIP	0.5412	0.5566	0.5489	0.3012	0.3192	0.3312	0.3121	0.3151	0.3386
	AHCH+BLIP	**0.6651**	**0.6612**	**0.6741**	0.3312	0.3351	0.3487	0.3512	0.3451	0.3572
	AHCH+SLIP	0.6241	0.6187	0.6455	0.3142	0.3287	0.3301	0.3547	0.3571	0.3789
	AHCH+CLIP	0.5815	0.5784	0.5861	**0.3792**	**0.3857**	**0.3821**	**0.4081**	0.3981	0.4202

achieves significant performance improvements across all datasets and various hash code lengths, validating the effectiveness of its key components. (2) Transformer-based frameworks (e.g., MITH, DSPH, and AHCH) generally outperform traditional CNN models (e.g., AGAH, DADH), highlighting their superiority in modeling complex sequential relationships. Compared to CNN-based baselines, the multimodal CLIP-based structure possesses stronger feature extraction capabilities, enabling the learning of more discriminative hash codes. (3) Although DDBH achieves sub-optimal performance, it lacks substantial improvements in the feature learning stage. In contrast, AHCH introduces a cue learning mechanism at this stage, thus consistently achieving the best results. On the small-scale MIRFLICKR25K dataset, AHCH improves the average performance by 3.74% in I2T and 4.86% in T2I retrieval tasks compared to the second-best method, fully demonstrating the effectiveness of cue learning in semantic alignment. (4) As shown in Table **??**, under the same experimental settings, AHCH achieves higher NDCG@1000 scores compared with existing VLMs methods in small-sample retrieval scenarios. This indicates that AHCH is able to rank relevant samples more accurately within the top 1000 retrieved results, thereby demonstrating a clear advantage in retrieval ranking quality. (5) On the large-scale NUSWIDE dataset and the MSCOCO dataset with long captions, AHCH still demonstrates strong performance in large-scale retrieval scenarios, both in terms of top-N accuracy (Fig. 4) and precision-recall (PR) metrics (Fig. 5). This performance gain is attributed to the Semantic Instance Alignment (SIA) component in the hash learning stage. SIA utilizes visual-text and text-visual cues to enhance complex feature representations, which optimizes the interactions across different modalities.

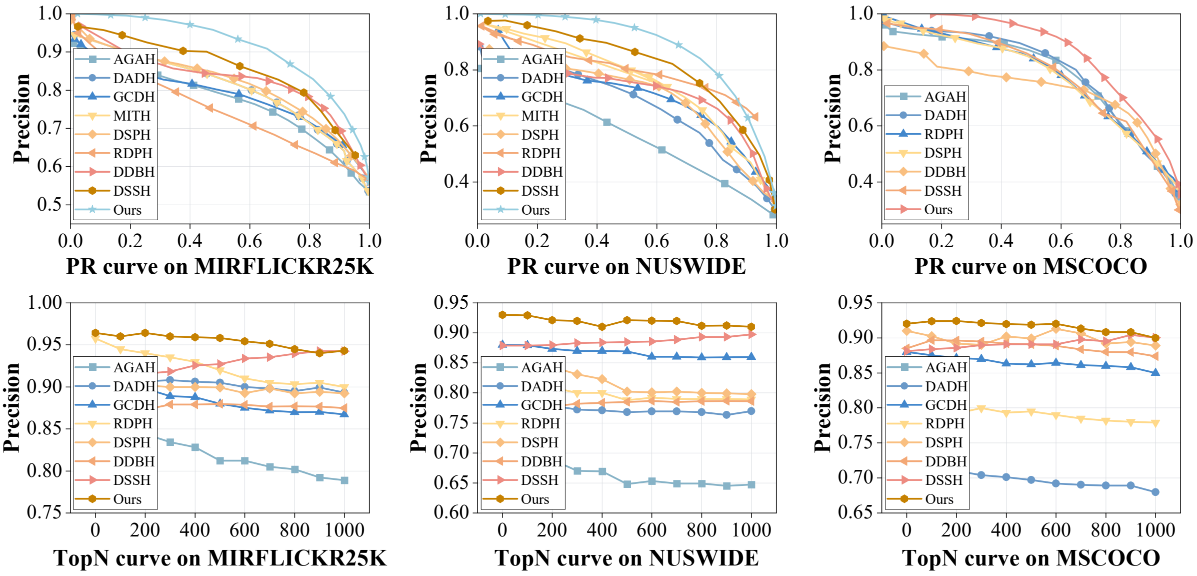

Fig. 5 T2I precision-recall and Top-N curves on MIRFLICKR25K, NUSWIDE, and MSCOCO at 64 bits.

Table 5 Ablation of cue learning and different fusion variants on MIRFLICKR25K at 32 and 64 bits.

Task	Method	mAP	
		32 bits	64 bits
I2T	BASE+Cue	0.8365	0.8447
	BASE+Cue+AHF	0.8718	0.8819
	BASE+Cue+SIA	0.8458	0.8570
	AHCH-WF	0.8781	0.8916
	AHCH-LF	**0.8841**	**0.8998**
T2I	BASE+Cue	0.8257	0.8314
	BASE+Cue+AHF	0.8569	0.8615
	BASE+Cue+SIA	0.8387	0.8462
	AHCH-WF	0.8583	0.8792
	AHCH-LF	**0.8619**	**0.8871**

Table 6 Comparison of different AHCH component configurations on MIRFLICKR25K at 32 and 64 bits.

Task	Method	mAP	
		32 bits	64 bits
I2T	BASE	0.8091	0.8291
	BASE+AHF	0.8593	0.8532
	BASE+SIA	0.8149	0.8187
	BASE+AHF+SIA	0.8612	0.8697
	AHCH	**0.8841**	**0.8998**
T2I	BASE	0.8123	0.8200
	BASE+AHF	0.8491	0.8372
	BASE+SIA	0.8139	0.8198
	BASE+AHF+SIA	0.8598	0.8680
	AHCH	**0.8619**	**0.8871**

4.4 Ablation Study

In this section, we systematically evaluate the contributions of each component on the MIRFLICKR25K dataset, involving five distinct learning configurations: (1) **BASE**, the basic model, which performs hash learning using only the Transformer-based CLIP encoder and the hash layer; (2) **BASE + AHF**, which incorporates Adaptive Hierarchical Fusion (AHF) strategy along with an image-caption branch to capture multi-level semantic interactions; (3) **BASE + SIA**, which augments the basic model with a Semantic Instance Alignment (SIA) module to enhance cross-modal content-based matching; (4) **BASE +AHF + SIA**, which combines both SIA and AHF to jointly model semantic alignment and hierarchical fusion; and (5) **AHCH**, the full model, which further integrates visual and textual cue learning strategies into the base framework, yielding an end-to-end cue-driven cross-modal hashing approach.

Ablation results are presented in Table 6, with analysis as follows. (1) The BASE model, which lacks structured cross-modal interaction and fine-grained alignment strategies, demonstrates suboptimal performance across experimental settings. This reflects limitations of the baseline method in cross-modal representation learning, largely attributable to its reliance on a static modeling strategy. (2) After incorporating Adaptive Hierarchical Fusion (AHF), inter-modal information interaction is enhanced, leading to an improvement in retrieval accuracy; the average mAP for I2T retrieval increases from 0.8169 to 0.8516, demonstrating the effectiveness of AHF in facilitating cross-modal interaction. (3) With the introduction of Semantic Instance Alignment (SIA), the learned real-valued hash codes exhibit a compact distribution and deliver stable retrieval performance with minor deviations across different code lengths.

Table 7 Ablation of SIA components on MIRFLICKR25K using fixed 64-bit hash codes in retrieval.

task	$\mathcal{J}_{inter}$	$\mathcal{J}_{intra}$	$\mathcal{J}_{quan}$	16bits	32bits	64bits	mean
I2T	✗	✗	✗	0.8121	0.8198	0.8251	0.8190
	✓	✗	✗	0.8219	0.8288	0.8329	0.8279
	✓	✓	✗	0.8355	0.8465	0.8511	0.8444
	✓	✓	✓	**0.8712**	**0.8841**	**0.8998**	**0.8850**
T2I	✗	✗	✗	0.8041	0.8121	0.8126	0.8096
	✓	✗	✗	0.8189	0.8276	0.8300	0.8255
	✓	✓	✗	0.8343	0.8455	0.8501	0.8433
	✓	✓	✓	**0.8675**	**0.8619**	**0.8871**	**0.8722**

Table 8 Efficiency comparison of visual backbones within AHCH using 64-bit hash codes on MIRFLICKR25K.

Visual Backbone	FLOPs (G)	Parameters (M)	Inference (ms)	Peak GPU Memory (GB)	mAP
ResNet-18	1.82	11.69	10.21	2.14	0.8722
ResNet-50	4.10	25.56	25.01	3.27	0.8812
ResNet-101	7.80	44.50	41.10	4.86	0.9081
CLIP ViT-B/32	4.41	88.22	25.41	3.92	**0.9198**

Specifically, in terms of average mAP, the performance difference between I2T and T2I retrieval is measured at only 0.0003 (0.8116 to 0.8119). (4) By integrating the cue-based optimization strategy, the AHCH framework achieves optimal performance across all code lengths. It not only improves retrieval accuracy but also yields more stable results under different configurations, validating the effectiveness of the proposed hierarchical cue design. (5) We evaluated various VLM variants using the NDCG@1000 metric on both known (D_k) and unknown (D_{unk}) retrieval databases, as shown in Table 4. On D_k, AHCH+BLIP outperformed AHCH+CLIP. The advantage of BLIP may be related to its more discriminative text representations learned through caption generation-based pre-training. However, for retrieval on D_{unk}, AHCH+CLIP achieved the best generalization performance, surpassing all other VLM variants. This is attributed to CLIP's symmetric image-text contrastive learning framework. Trained on ultra-large-scale noisy web data, CLIP learns broad and robust semantic alignment between images and texts, rather than capturing dataset-specific idiosyncrasies. Therefore, when encountering out-of-distribution unknown data, CLIP as the backbone provides more generalized feature representations for hash learning, enabling the AHCH framework to exhibit stronger cross-domain generalization capability.

4.5 Algorithm Efficiency

To evaluate the efficiency of the proposed model, we compare the training and encoding time of several methods on MIRFLICKR25K and NUSWIDE dataset with 64bits hash codes, as shown in Fig. 6. The results demonstrate that the network architecture greatly influences both training efficiency and retrieval performance. SSAH and AGAH employ shallow CNNs and lightweight MLPs for feature extraction, achieving faster training due to their simple architectures. In contrast, Transformer-based methods such as DSPH and DDBH rely on large-scale VLMs, which enhance retrieval accuracy but introduce significant computational overhead. Moreover, we assess the inference time for image-to-text retrieval on the MIRFLICKR25K dataset, as reported in Table 8. With 64bits hash codes and 512-dimensional features, the analysis shows that inference time is mainly determined by FLOPs rather than the number of parameters. Although our method involves more parameters, its FLOPs are comparable to ResNet-50, resulting in similar inference latency while achieving superior retrieval performance.

From the viewpoint of time complexity, the offline construction of the hash database in AHCH scales linearly with the number of multimedia instances. Let $N_g = |D_k| + |D_{\text{unk}}|$ denote the total number of image-text pairs in the gallery. Since each instance requires one forward pass through the frozen CLIP encoders and hash heads to generate a K-bit code, the overall database construction cost is $\mathcal{O}(N_g C_{\text{enc}} + N_g K) \approx \mathcal{O}(N_g C_{\text{enc}})$, where C_{enc} denotes the FLOPs of a single encoder pass.

4.6 Parameter Sensitivity

To investigate the parameter sensitivity of the Semantic Instance Alignment (SIA) component in the AHCH framework, we conducted an empirical study on MIRFLICKR25K and NUSWIDE dataset with a

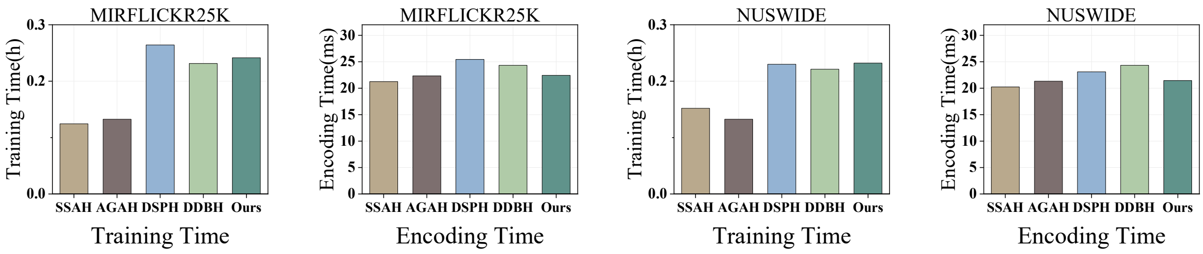

Fig. 6 Training and encoding time of our AHCH on MIRFLICKR25K under the same hardware settings.

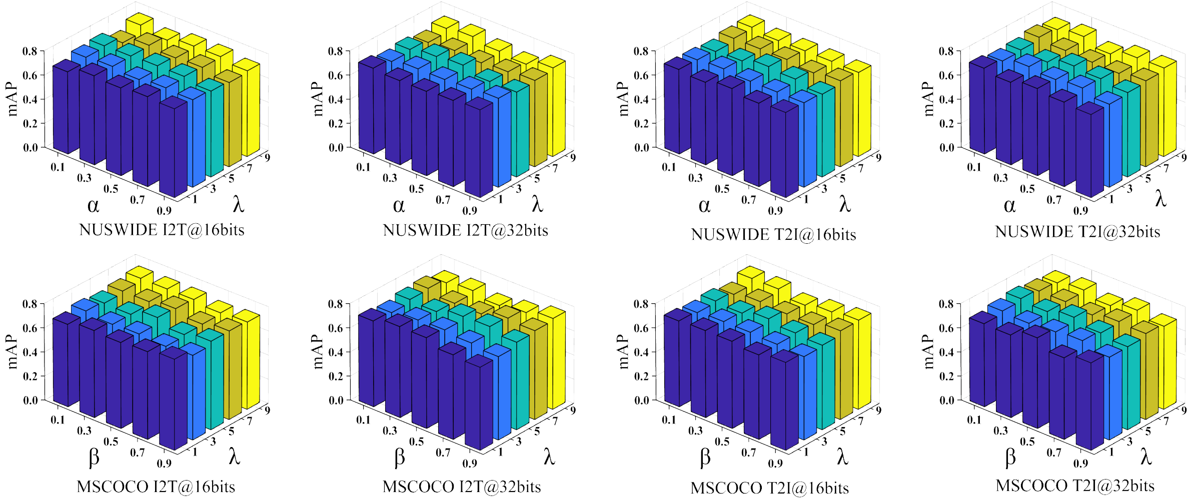

Fig. 7 mAP sensitivity to α, β, and λ on NUSWIDE and MSCOCO at 16 and 32 bits for both directions.

hash code length of 64 bits. Following Eq. (23), we examined three hyperparameters (α, β, λ) by varying each individually while keeping the others fixed. As shown in Fig. 7, the model performance fluctuates with different parameter settings, which also validates the contribution of each component. Based on these observations, we conclude that the AHCH framework achieves satisfactory performance when the parameters are set to $\{\alpha = 0.5, \beta = 0.5, \lambda = 5\}$. When the parameter settings vary, the model performance changes accordingly. The hyperparameters α and β primarily regulate the discrepancy between long- and short-sequence feature mappings within and across modalities, i.e., (f^{pv}, f^{ev}). If α and β are set too large, or if their values differ significantly, the stability of feature alignment will be affected, thereby weakening semantic consistency. Moreover, we further examined the influence of λ, which controls the strength of the quantization regularization term. When λ is too small (e.g., $\lambda \leq 5$), the real-valued hash features deviate from the binary codes, resulting in increased quantization error and a noticeable drop in retrieval performance. Conversely, an excessively large λ (e.g., $\lambda \geq 5$) over-constrains the binarization process and weakens semantic alignment across modalities.

Considering the internal balance of SIA components, we conducted a comprehensive analysis as shown in Table 7. Starting from a variant without these losses, introducing $\mathcal{L}_{inter}$ consistently improves both I2T and T2I retrieval, confirming the importance of explicit cross-modal alignment. Adding $\mathcal{L}_{intra}$ on top of $\mathcal{L}_{inter}$ brings further gains by better preserving semantic structure within each modality. The best performance is achieved when $\mathcal{L}_{quan}$ is also enabled, demonstrating that a balanced combination of inter-modal alignment, intra-modal preservation, and quantization is crucial for high-quality hash codes.

4.7 Visualization Results

(1) qualitative analysis: We further conduct a qualitative analysis of the embedding space using four groups of animal examples, denoted as Easy, Hard, Near and Far (Fig. 8). Each group consists of four image-text pairs. The Easy and Hard groups contain images from the same category (cat), where Easy uses visually similar instances while Hard exhibits large variations in pose, illumination and background. Near and Far are composed of different animal categories that are semantically close (all pets) and clearly different, respectively. For each group, we visualize the 4×4 cosine similarity matrix between

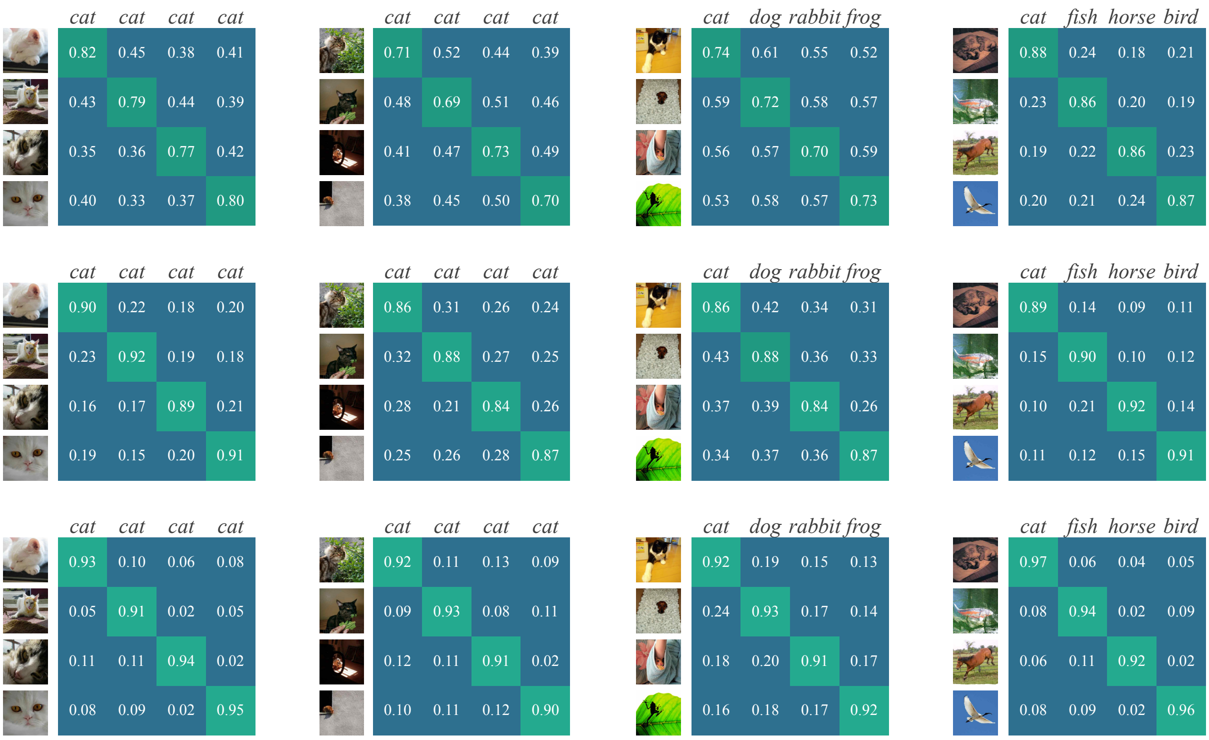

Fig. 8 Qualitative visualization on MIRFLICKR25K from Eq. 14. Columns correspond to four animal groups: Easy, Hard, Near, and Far. Rows correspond to three model variants: the baseline without AHF denoted as Origin, AHCH-WF, and AHCH-LF. Colors indicate cosine similarities between image embeddings as rows and text queries as columns.

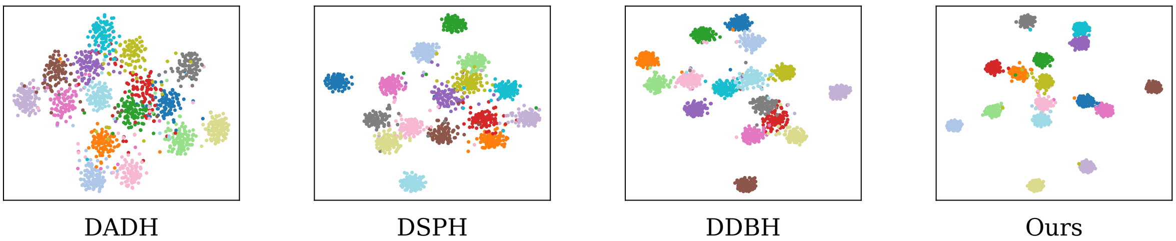

Fig. 9 t-SNE visualization of the learned image hash-code representations on MIRFLICKR25K under 16bits length.

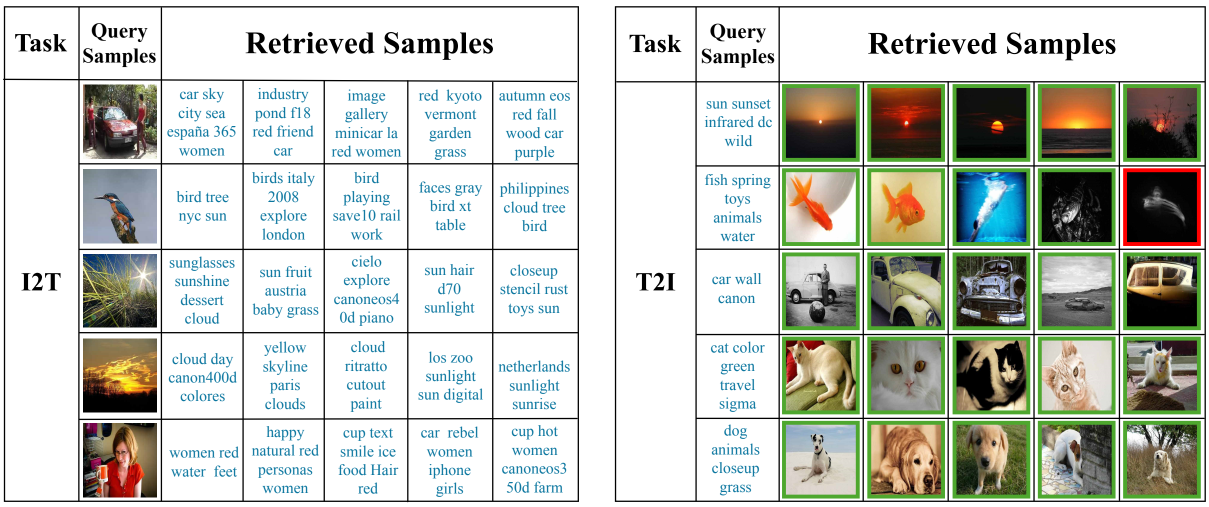

Fig. 10 Top-5 image-text retrieval results on MIRFLICKR25K using 64-bit hash codes for qualitative comparison across different cross-modal retrieval methods. Green and red markers denote relevant and irrelevant samples, respectively.

image and text embeddings. The baseline model without AHF (Origin, first row) produces relatively blurred similarity patterns: diagonal entries (matched pairs) are not always dominant and many off-diagonal elements still have high responses, especially in the Hard and Near groups. In contrast, the two AHF variants, AHCH-WF and AHCH-LF, substantially increase the diagonal similarities and suppress the off-diagonal ones. In particular, AHCH-LF (third row) yields the most discriminative matrices, with consistently high diagonal values and almost negligible off-diagonal responses across all four groups. These observations confirm that the proposed AHF, especially the learnable fusion branch, effectively enhances intra-class consistency and reduces cross-class confusion in the cross-modal embedding space, which explains the performance gains observed in the quantitative results.

(2) t-SNE visualization: To intuitively evaluate the discriminative ability of the learned hash codes, we employ t-distributed Stochastic Neighbor Embedding [49] to project high-dimensional embeddings into a two-dimensional space. As illustrated in Fig. 10, we visualize the embedding results of image hash codes and text hash codes on the MSCOCO dataset. It can be observed that the clusters generated by AHCH are more compact within classes and more clearly separated across classes, demonstrating stronger positive neighbor aggregation and negative neighbor separation. This improvement can be attributed to two key factors: (1) the bidirectional flow of relational information, which effectively alleviates the impact of imbalanced samples, and (2) the adoption of an alternating quantization strategy, which jointly optimizes embedding error and quantization error. Consequently, the generated hash codes achieve significant enhancements in both the consistency and the discriminability of the generated hash codes.

(3) Top-5 Retrieval Results: Most existing methods in cross-modal hashing primarily evaluate retrieval performance, without directly examining the model's effectiveness on downstream tasks. In practical applications, however, the performance of downstream tasks is of greater concern. It remains unclear whether improvements in VLMs retrieval performance can be directly translated into tangible gains for downstream tasks. Although this work mainly reports retrieval performance, we conducted a small-scale downstream evaluation using the improved retrieval results for a simple image-text matching test to preliminarily assess the transferability of retrieval improvements. As shown in Fig. 9, considering the top-5 image-text retrieval results on the MIRFLICKR25K dataset with 64bits hash codes, the samples retrieved by AHCH consistently maintain highly relevant to the queries, indicating that the generated binary codes effectively capture the underlying cross-modal relationships.

5 Conclusion

In this paper, we have proposed Adaptive Hierarchical Cue Hashing (AHCH), a novel framework for cross-modal retrieval. AHCH enhances semantic consistency and discriminability by introducing a cue-guided mechanism that dynamically strengthens feature representations of both modalities. It further adopts an adaptive hierarchical fusion strategy to integrate multi-level features, preserving fine-grained content information while mitigating inter-modal discrepancies. Additionally, by optimizing instance-level similarity, the framework not only ensures precise cross-modal alignment but also generates compact, high-quality hash codes. Extensive experiments on three benchmark datasets demonstrate that AHCH achieves consistent and competitive improvements over strong baselines in retrieval accuracy and generalization across various scenarios. Beyond conventional retrieval tasks, integrated image-based deep learning and language models have demonstrated practical value in specialized domains such as primary healthcare, illustrating the broader application potential of joint visual and textual modeling [50].

Despite these encouraging results, AHCH has limitations. The current method and evaluation mainly target image-text retrieval, and its generalization to other modalities requires further verification. Moreover, experiments are conducted on three public benchmarks, which may not fully reflect the diversity of real-world scenarios or severe domain shifts. Finally, the cue-guided and hierarchical fusion designs add optimization complexity, and performance may depend on hyperparameter choices such as hash length. Future work will extend AHCH to broader modalities and explore more efficient optimization strategies.

Author Contributions Y.W. and M.W. conceived the study and designed the experiments. Y.W. developed the methodology and performed the formal analysis. M.W. conducted the investigation, curated the data, and prepared the visualizations. Both authors wrote, reviewed, and edited the manuscript.

Data Availability The code, data, and results supporting this study have been uploaded to GitHub and can be accessed via the following link for reproducibility: https://github.com/wyl20000205/AHCH.

Declarations

Conflict of interest The authors declare no conflict of interest.

References

[1] Liu, Z., Chen, F., Xu, J., Pei, W., Lu, G.: Image-text retrieval with cross-modal semantic importance consistency. IEEE Transactions on Circuits and Systems for Video Technology **33**(5), 2465–2476 (2023) https://doi.org/10.1109/TCSVT.2022.3220297

[2] Wang, Z., Gao, Z., Han, M., Yang, Y., Shen, H.T.: Estimating the semantics via sector embedding for image-text retrieval. IEEE Transactions on Multimedia **26**, 10342–10353 (2024) https://doi.org/10.1109/TMM.2024.3407664

[3] Ma, Y., Wang, M., Lu, G., Sun, Y.: Multi-label semantic sharing based on graph convolutional network for image-to-text retrieval. The Visual Computer **41**(3), 1827–1840 (2025) https://doi.org/10.1007/s00371-024-03496-y

[4] Zhu, M., Zhao, C., Wang, N., Gu, F., Liu, Y., Li, X.: Domain-aware double attention network for zero-shot sketch-based image retrieval with similarity loss. The Visual Computer **40**(5), 3091–3101 (2024) https://doi.org/10.1007/s00371-023-03012-8

[5] Cao, Y., Liu, B., Long, M., Wang, J.: Cross-modal hamming hashing. In: Ferrari, V., Hebert, M., Sminchisescu, C., Weiss, Y. (eds.) Computer Vision – ECCV 2018, pp. 207–223. Springer, Cham (2018)

[6] Su, H., Han, M., Liang, J., Liang, J., Yu, S.: Deep supervised hashing with hard example pairs optimization for image retrieval. The Visual Computer **39**(11), 5405–5420 (2023) https://doi.org/10.1007/s00371-022-02668-y

[7] Wu, B., Yang, Q., Zheng, W.-S., Wang, Y., Wang, J.: Quantized correlation hashing for fast cross-modal search. In: Proceedings of the 24th International Conference on Artificial Intelligence. IJCAI'15, pp. 3946–3952. AAAI Press, ??? (2015)

[8] Wang, D., Gao, X., Wang, X., He, L., Yuan, B.: Multimodal discriminative binary embedding for large-scale cross-modal retrieval. IEEE Transactions on Image Processing **25**(10), 4540–4554 (2016) https://doi.org/10.1109/TIP.2016.2592800

[9] Mahalle, V.S., Kandoi, N.M., Patil, S.B.: A powerful method for interactive content-based image retrieval by variable compressed convolutional info neural networks. The Visual Computer **40**(8), 5259–5285 (2024) https://doi.org/10.1007/s00371-023-03104-5

[10] Liu, Q., Ge, Y., Wang, S., Wang, T., Xu, J.: Dynamic manifold-based sample selection in contrastive learning for remote sensing image retrieval. The Visual Computer **41**(6), 4111–4127 (2025) https://doi.org/10.1007/s00371-024-03649-z

[11] Gu, W., Gu, X., Gu, J., Li, B., Xiong, Z., Wang, W.: Adversary guided asymmetric hashing for cross-modal retrieval. In: Proceedings of the 2019 on International Conference on Multimedia Retrieval. ICMR '19, pp. 159–167. Association for Computing Machinery, New York, NY, USA (2019). https://doi.org/10.1145/3323873.3325045 . https://doi.org/10.1145/3323873.3325045

[12] Bai, C., Zeng, C., Ma, Q., Zhang, J., Chen, S.: Deep adversarial discrete hashing for cross-modal retrieval. In: Proceedings of the 2020 International Conference on Multimedia Retrieval. ICMR '20, pp. 525–531. Association for Computing Machinery, New York, NY, USA (2020). https://doi.org/10.1145/3372278.3390711 . https://doi.org/10.1145/3372278.3390711

[13] Li, J., Tang, Y.: Vit-sir: vision transformer-based shoe image retrieval with enhanced feature representation. The Visual Computer **41**(11), 9057–9075 (2025) https://doi.org/10.1007/s00371-025-03913-w

[14] Huo, Y., Qin, Q., Zhang, W., Huang, L., Nie, J.: Deep hierarchy-aware proxy hashing with self-paced learning for cross-modal retrieval. IEEE Transactions on Knowledge and Data Engineering **36**(11), 5926–5939 (2024) https://doi.org/10.1109/TKDE.2024.3401050

[15] Wang, G., Cheng, S., Du, A., Zou, Q.: Covariance attention guidance mamba hashing for cross-modal retrieval. Engineering Applications of Artificial Intelligence **152**, 110777 (2025) https://doi.org/10.1016/j.engappai.2025.110777

[16] Al-Jebrni, A.H., Jiang, Z., Qin, Y., Qiu, Y., Jiang, N., Li, G., Zheng, Y.: Visual-language foundation models and artificial intelligence agents in healthcare: Bridging from technological innovation to clinical impact. Computational Visual Media, 1–38 (2026) https://doi.org/10.26599/CVM.2026.9450530 . Early Access

[17] Zhu, C., Hu, W., Hou, J., Qin, Q., Zhang, W., Huang, L.: Deep adaptive gradient-triplet hashing

for cross-modal retrieval. Expert Systems with Applications **291**, 128566 (2025) https://doi.org/10.1016/j.eswa.2025.128566
[18] Radford, A., Kim, J.W., Hallacy, C., Ramesh, A., Goh, G., Agarwal, S., Sastry, G., Askell, A., Mishkin, P., Clark, J., Krueger, G., Sutskever, I.: Learning transferable visual models from natural language supervision. In: Meila, M., Zhang, T. (eds.) Proceedings of the 38th International Conference on Machine Learning, ICML 2021, 18-24 July 2021, Virtual Event. Proceedings of Machine Learning Research, vol. 139, pp. 8748–8763. PMLR, ??? (2021). http://proceedings.mlr.press/v139/radford21a.html
[19] Zhang, Z., Guo, X.-G., Zhuo, J., Ma, H.: Two-stage fine-tuning clip by introducing structure knowledge for few-shot classification. The Visual Computer **41**(13), 10977–10986 (2025) https://doi.org/10.1007/s00371-025-04080-8
[20] Qin, Q., Huo, Y., Zhang, W., Huang, L., Nie, J.: Deep discriminative boundary hashing for cross-modal retrieval. IEEE Transactions on Circuits and Systems for Video Technology **35**(10), 10557–10570 (2025) https://doi.org/10.1109/TCSVT.2025.3570128
[21] Zhang, Q., Zhao, J.-R., Liu, X.-Q., Zhan, Y.-W., Chen, Z.-D., Luo, X., Xu, X.-S.: Hypergraph-based clip hashing for unsupervised cross-modal retrieval. Knowledge-Based Systems **330**, 114508 (2025) https://doi.org/10.1016/j.knosys.2025.114508
[22] Li, M., Wang, H.: Unsupervised deep cross-modal hashing by knowledge distillation for large-scale cross-modal retrieval. In: Proceedings of the 2021 International Conference on Multimedia Retrieval. ICMR '21, pp. 183–191. Association for Computing Machinery, New York, NY, USA (2021). https://doi.org/10.1145/3460426.3463626 . https://doi.org/10.1145/3460426.3463626
[23] Zhu, L., Wu, X., Li, J., Zhang, Z., Guan, W., Shen, H.T.: Work together: Correlation-identity reconstruction hashing for unsupervised cross-modal retrieval. IEEE Transactions on Knowledge and Data Engineering **35**(9), 8838–8851 (2023) https://doi.org/10.1109/TKDE.2022.3218656
[24] Yang, Z., Yang, L., Raymond, O.I., Zhu, L., Huang, W., Liao, Z., Long, J.: Nsdh: A nonlinear supervised discrete hashing framework for large-scale cross-modal retrieval. Knowledge-Based Systems **217**, 106818 (2021) https://doi.org/10.1016/j.knosys.2021.106818
[25] Qin, Q., Ge, M., Zhang, W., Huang, L., Nie, J.: Deep stochastic spherical hashing with von mises-fisher distributions for cross-modal retrieval. IEEE Transactions on Knowledge and Data Engineering **38**(5), 2655–2669 (2026) https://doi.org/10.1109/TKDE.2026.3665949
[26] Hu, H., Xie, L., Hong, R., Tian, Q.: Creating something from nothing: Unsupervised knowledge distillation for cross-modal hashing. In: 2020 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), pp. 3120–3129 (2020). https://doi.org/10.1109/CVPR42600.2020.00319
[27] Shi, Y., Zhao, Y., Liu, X., Zheng, F., Ou, W., You, X., Peng, Q.: Deep adaptively-enhanced hashing with discriminative similarity guidance for unsupervised cross-modal retrieval. IEEE Transactions on Circuits and Systems for Video Technology **32**(10), 7255–7268 (2022) https://doi.org/10.1109/TCSVT.2022.3172716
[28] Li, J., Wong, W.K., Jiang, L., Fang, X., Xie, S., Xu, Y.: Ckdh: Clip-based knowledge distillation hashing for cross-modal retrieval. IEEE Transactions on Circuits and Systems for Video Technology **34**(7), 6530–6541 (2024) https://doi.org/10.1109/TCSVT.2024.3350695
[29] Tu, R.-C., Jiang, J., Lin, Q., Cai, C., Tian, S., Wang, H., Liu, W.: Unsupervised cross-modal hashing with modality-interaction. IEEE Transactions on Circuits and Systems for Video Technology **33**(9), 5296–5308 (2023) https://doi.org/10.1109/TCSVT.2023.3251395
[30] Chen, Y., Yang, Z., Long, J.: Unsupervised hierarchical dynamic similarity hashing for multimedia retrieval. In: ICASSP 2025 - 2025 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), pp. 1–5 (2025). https://doi.org/10.1109/ICASSP49660.2025.10889267
[31] Xiong, M., Hu, K., Wang, Z., Hu, R., Muhammad, K., Del Ser, J., Yang, X., Sheng, B.: Adaptive clustering and weighted regularization contrastive learning framework for unsupervised person re-identification. IEEE Transactions on Multimedia **27**, 6896–6907 (2025) https://doi.org/10.1109/TMM.2025.3590938
[32] Zhang, J., Dai, L., Ye, F., Chen, Z., Li, P., Yang, X., Sheng, B.: Dataset distillation via a noise-unconstrained generative model. IEEE Transactions on Pattern Analysis and Machine Intelligence **48**(9), 10809–10826 (2026) https://doi.org/10.1109/TPAMI.2026.3690778
[33] Peng, S.-J., Jiang, X., Liu, X., Du, J.-X., Cao, J.: Lpgoh: Label-prototype guided online hashing for efficient cross-modal retrieval. Knowledge-Based Systems **327**, 114159 (2025) https://doi.org/10.1016/j.knosys.2025.114159
[34] Devlin, J., Chang, M.-W., Lee, K., Toutanova, K.: BERT: Pre-training of deep bidirectional transformers for language understanding. In: Burstein, J., Doran, C., Solorio, T. (eds.) Proceedings of the

2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers), pp. 4171–4186. Association for Computational Linguistics, Minneapolis, Minnesota (2019). https://doi.org/10.18653/v1/N19-1423 . https://aclanthology.org/N19-1423/
[35] Radford, A., Wu, J., Child, R., Luan, D., Amodei, D., Sutskever, I.: Language models are unsupervised multitask learners. OpenAI (2019). Accessed: 2024-11-15
[36] Chen, S., Zhong, X., Yang, X., Yang, L., Sheng, B., Kot, A.C.: Fine-grained lexical-centric semantic network for coherent video paragraph captioning. IEEE Transactions on Multimedia **28**, 6840–6851 (2026) https://doi.org/10.1109/TMM.2026.3668632
[37] Shao, Z., Yang, H., Zhu, H., Yao, R., Zou, L., Li, M., Sheng, B.: Spatio-temporal disentanglement and constrained self-attention for multi-modal deception detection. ACM Transactions on Multimedia Computing, Communications, and Applications **22**(2) (2026) https://doi.org/10.1145/3783994
[38] Huiskes, M.J., Lew, M.S.: The mir flickr retrieval evaluation. In: Proceedings of the 1st ACM International Conference on Multimedia Information Retrieval. MIR '08, pp. 39–43. Association for Computing Machinery, New York, NY, USA (2008). https://doi.org/10.1145/1460096.1460104 . https://doi.org/10.1145/1460096.1460104
[39] Chua, T.-S., Tang, J., Hong, R., Li, H., Luo, Z., Zheng, Y.: Nus-wide: a real-world web image database from national university of singapore. In: Proceedings of the ACM International Conference on Image and Video Retrieval. CIVR '09. Association for Computing Machinery, New York, NY, USA (2009). https://doi.org/10.1145/1646396.1646452 . https://doi.org/10.1145/1646396.1646452
[40] Lin, T.-Y., Maire, M., Belongie, S., Hays, J., Perona, P., Ramanan, D., Dollár, P., Zitnick, C.L.: Microsoft coco: Common objects in context. In: Fleet, D., Pajdla, T., Schiele, B., Tuytelaars, T. (eds.) Computer Vision – ECCV 2014, pp. 740–755. Springer, Cham (2014)
[41] Sennrich, R., Haddow, B., Birch, A.: Neural machine translation of rare words with subword units. In: Erk, K., Smith, N.A. (eds.) Proceedings of the 54th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), pp. 1715–1725. Association for Computational Linguistics, Berlin, Germany (2016). https://doi.org/10.18653/v1/P16-1162 . https://aclanthology.org/P16-1162/
[42] Bai, C., Zeng, C., Ma, Q., Zhang, J.: Graph convolutional network discrete hashing for cross-modal retrieval. IEEE Transactions on Neural Networks and Learning Systems **35**(4), 4756–4767 (2024) https://doi.org/10.1109/TNNLS.2022.3174970
[43] Liu, Y., Wu, Q., Zhang, Z., Zhang, J., Lu, G.: Multi-granularity interactive transformer hashing for cross-modal retrieval. In: Proceedings of the 31st ACM International Conference on Multimedia. MM '23, pp. 893–902. Association for Computing Machinery, New York, NY, USA (2023). https://doi.org/10.1145/3581783.3612411 . https://doi.org/10.1145/3581783.3612411
[44] Huo, Y., Qin, Q., Dai, J., Wang, L., Zhang, W., Huang, L., Wang, C.: Deep semantic-aware proxy hashing for multi-label cross-modal retrieval. IEEE Transactions on Circuits and Systems for Video Technology **34**(1), 576–589 (2024) https://doi.org/10.1109/TCSVT.2023.3285266
[45] Song, G., Huang, K., Su, H., Song, F., Yang, M.: Deep ranking distribution preserving hashing for robust multi-label cross-modal retrieval. IEEE Transactions on Multimedia **26**, 7027–7042 (2024) https://doi.org/10.1109/TMM.2024.3358995
[46] Yang, K., Deng, J., An, X., Li, J., Feng, Z., Guo, J., Yang, J., Liu, T.: Alip: Adaptive language-image pre-training with synthetic caption. In: 2023 IEEE/CVF International Conference on Computer Vision (ICCV), pp. 2910–2919 (2023). https://doi.org/10.1109/ICCV51070.2023.00273
[47] Li, J., Li, D., Xiong, C., Hoi, S.: Blip: Bootstrapping language-image pre-training for unified vision-language understanding and generation. In: ICML (2022)
[48] Mu, N., Kirillov, A., Wagner, D., Xie, S.: Slip: Self-supervision meets language-image pre-training. In: Avidan, S., Brostow, G., Cissé, M., Farinella, G.M., Hassner, T. (eds.) Computer Vision – ECCV 2022, pp. 529–544. Springer, Cham (2022)
[49] Maaten, L., Hinton, G.: Visualizing data using t-sne. Journal of Machine Learning Research **9**(86), 2579–2605 (2008)
[50] Li, J., *et al.*: Integrated image-based deep learning and language models for primary diabetes care. Nature Medicine **30**(10), 2886–2896 (2024) https://doi.org/10.1038/s41591-024-03139-8